
RPX: A Unified Real-World Benchmark for Evaluating Robot Perception Under Embodied Deployment Conditions

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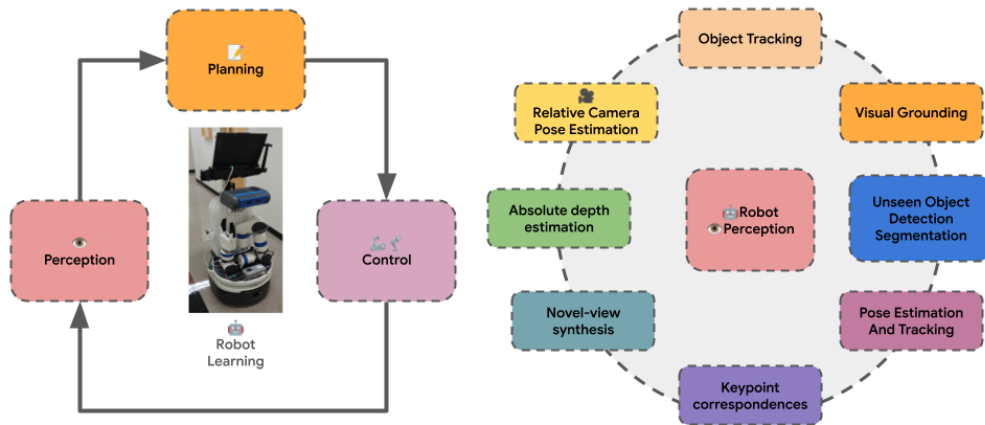


Figure 1: RPX Framework Overview.

Abstract

1

2 1 Introduction

3 Recent advances in foundation vision and vision-language models have significantly influenced
4 modern robot manipulation systems. Perception modules for segmentation, tracking, depth estimation,
5 and grounding are increasingly instantiated using large pretrained backbones. While these models
6 achieve impressive performance on standard vision benchmarks, they are typically evaluated in
7 isolation on task-specific datasets that differ in object categories, scene composition, resolution, and
8 camera motion characteristics. These benchmarks rarely reflect embodied deployment conditions,
9 where perception operates under continuous motion, clutter, human interaction, varying zoom, motion
10 blur, and real-world lighting variability.

11 Robotic perception, however, is inherently embodied. A mobile platform continuously changes
12 viewpoint while interacting with cluttered environments. Predictions must remain stable under
13 motion, robust during interaction-induced occlusion, and reliable across scene reconfiguration. In
14 practice, deployment failures often arise not from poor static accuracy, but from instability during
15 camera motion, degradation under occlusion, or fragility in difficult scenes. Yet existing benchmarks

do not provide a unified framework to evaluate perception modules under these deployment-relevant conditions.

To address this gap, we introduce **RPX**, a unified real-world benchmark designed specifically for evaluating robot perception under embodied deployment conditions. RPX comprises 75,000 RGB-D frames across 100 scenes captured using robot-grade sensors (Intel D435 and T265) under human-driven trajectories that emulate mobile robot exploration. Each scene follows a structured protocol consisting of three phases: (i) cluttered configuration, (ii) human interaction, and (iii) post-interaction clean configuration. This design enables controlled evaluation of robustness to occlusion, motion blur, object rearrangement, and scene transitions.

Unlike existing benchmarks that evaluate segmentation, depth, tracking, or grounding independently on separate datasets, RPX supports multi-task evaluation on identical scenes and trajectories. This unified structure enables consistent comparison of perception modules under shared environmental and motion conditions. We provide verified instance masks, tracklets, monocular depth supervision, ground-truth camera pose, and language annotations, enabling comprehensive evaluation across perception tasks central to robot manipulation.

To further characterize perceptual ambiguity, we introduce **Effort-Stratified Difficulty (ESD)** splits derived from annotation refinement statistics. Instances requiring greater refinement effort correspond to increased occlusion, clutter, thin structures, and motion complexity. This stratification enables controlled robustness analysis across easy, medium, and hard subsets.

To evaluate deployment readiness, we report task-specific performance across scene phase (clutter / interaction / clean) and difficulty level (easy / medium / hard), and introduce weighted robustness summaries emphasizing challenging conditions. In addition to perceptual robustness, we measure computational efficiency including inference latency and memory footprint to assess practical deployability. We benchmark 10 representative open- and closed-source foundation models per task, revealing substantial discrepancies between static benchmark accuracy, embodied robustness, and computational efficiency under real-world conditions. RPX establishes a standardized testbed and toolkit for evaluating robot perception models in deployment-relevant settings and facilitates principled model selection for downstream manipulation systems.

In practical robotic systems, perceptual robustness must be balanced with computational efficiency. High latency or excessive memory consumption can limit real-time deployment on resource-constrained platforms. Accordingly, RPX reports inference latency and peak memory usage alongside perceptual metrics, enabling holistic evaluation of deployment readiness.

To ensure reproducibility and accessibility, we release RPX with a pip-installable toolkit providing standardized splits, data loaders, metric implementations, and adapter interfaces for rapid model integration. RPX establishes a standardized real-world testbed for evaluating robot perception models in embodied settings and supports principled selection of perceptual backbones for downstream manipulation pipelines.

Taking real world robotics scenarios as reference from FewSOL, Proto-CLIP, NIDS-Net, SceneReplica, AutoSem iTteach, HRT1, RPX tries to evaluate perception models in a more organized and unified way.

2 Related Work

Large models. Benchmarks in robotics. P et al. [2023, 2024a,b], Khargonkar et al. [2024], Lu et al. [2025], Allu* et al. [2025]

59 3 RPX Difficulty Features

60 Given a scene s and phase p , we define a video sequence of T frames:

$$\mathcal{V}_{s,p} = \{I_t, D_t, M_t, \mathbf{T}_t\}_{t=1}^T$$

61 where I_t is the RGB image, D_t is the depth map, M_t is the instance mask, and $\mathbf{T}_t \in SE(3)$ is the
62 camera pose.

63 We extract a set of features $\{f_i\}$ capturing annotation effort, scene complexity, sensing quality,
64 temporal stability, and camera motion.

65 3.1 Annotation Effort

66 Let $\mathcal{K}_{s,p} = \{k_j\}$ denote the number of mask refinement iterations.

$$f_{\text{iter-mean}} = \frac{1}{|\mathcal{K}_{s,p}|} \sum_j k_j \quad (1)$$

$$f_{\text{iter-max}} = \max_j k_j \quad (2)$$

67 3.2 Scene Complexity and Occlusion

68 Let $\mathcal{O}_t$ be the set of object instances in frame t .

Object Count

$$f_{\text{obj-mean}} = \frac{1}{T} \sum_{t=1}^T |\mathcal{O}_t| \quad (3)$$

$$f_{\text{obj-std}} = \sqrt{\frac{1}{T} \sum_{t=1}^T (|\mathcal{O}_t| - f_{\text{obj-mean}})^2} \quad (4)$$

69 **Occlusion** Let $\text{bbox}(o)$ denote the bounding box of object o . The occlusion of object o is:

$$\text{occ}(o) = \frac{\sum_{o' \neq o} \text{area}(\text{bbox}(o) \cap \text{bbox}(o'))}{\text{area}(\text{bbox}(o))}$$

70 Then:

$$f_{\text{occ-mean}} = \mathbb{E}_o[\text{occ}(o)] \quad (5)$$

$$f_{\text{occ-p90}} = \text{Percentile}_{90}(\text{occ}(o)) \quad (6)$$

$$f_{\text{occ-heavy}} = \frac{1}{|\mathcal{O}|} \sum_o \mathbb{1}[\text{occ}(o) > \tau] \quad (7)$$

71 3.3 Depth and Sensing Quality

Invalid Depth Ratio

$$f_{\text{depth-invalid}} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_x \mathbb{1}[D_t(x) \leq 0 \vee \text{NaN}]}{|\Omega|} \quad (8)$$

$$f_{\text{depth-invalid-mask}} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_{x \in M_t} \mathbb{1}[D_t(x) \leq 0 \vee \text{NaN}]}{|M_t|} \quad (9)$$

Depth Variability

$$f_{\text{depth-std}} = \frac{1}{T} \sum_{t=1}^T \text{Std}(D_t(x)), \quad x \in \text{valid pixels} \quad (10)$$

72 **Photometric-Depth Conflict** Let $I_t(x)$ denote grayscale intensity.

$$f_{\text{specular}} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_x \mathbb{I}[I_t(x) > \tau_b \wedge D_t(x) \text{ invalid}]}{|\Omega|} \quad (11)$$

$$f_{\text{dark-conflict}} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_x \mathbb{I}[I_t(x) < \tau_d \wedge D_t(x) \text{ invalid}]}{|\Omega|} \quad (12)$$

73 3.4 Temporal Stability

74 Let A_t^o be the mask area of object o at frame t .

Mask Area Variability

$$\text{CV}(o) = \frac{\sigma(A^o)}{\mu(A^o)} \quad (13)$$

$$f_{\text{area-cv}} = \mathbb{E}_o[\text{CV}(o)] \quad (14)$$

Area Drop Events

$$f_{\text{area-drop}} = \frac{1}{T} \sum_{t=2}^T \mathbb{I}\left[\frac{A_{t-1}^o - A_t^o}{A_{t-1}^o} > \delta\right] \quad (15)$$

75 **Visibility Instability** Let $v_t^o \in \{0, 1\}$ indicate visibility:

$$f_{\text{vis-instability}} = \frac{1}{T} \sum_{t=2}^T \mathbb{I}[v_t^o \neq v_{t-1}^o] \quad (16)$$

76 3.5 Camera Motion

77 Let $\mathbf{t}_t$ and $\mathbf{R}_t$ denote translation and rotation.

$$\Delta \mathbf{t}_t = \|\mathbf{t}_{t+1} - \mathbf{t}_t\| \quad (17)$$

$$\Delta \mathbf{R}_t = \text{angle}(\mathbf{R}_t^{-1} \mathbf{R}_{t+1}) \quad (18)$$

$$f_{\text{trans-mean}} = \mathbb{E}[\Delta \mathbf{t}_t] \quad (19)$$

$$f_{\text{trans-p90}} = \text{Percentile}_{90}(\Delta \mathbf{t}_t) \quad (20)$$

$$f_{\text{rot-mean}} = \mathbb{E}[\Delta \mathbf{R}_t] \quad (21)$$

$$f_{\text{rot-p90}} = \text{Percentile}_{90}(\Delta \mathbf{R}_t) \quad (22)$$

Motion Jerk

$$f_{\text{jerk}} = \mathbb{E}[|\Delta \mathbf{t}_{t+1} - \Delta \mathbf{t}_t|] \quad (23)$$

78 3.6 Object Consistency

$$f_{\text{obj}} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}[|\mathcal{O}_t| = 7] \quad (24)$$

79 3.7 Fisheye Features (Optional)

80 Let F_t^L, F_t^R denote left and right fisheye images.

$$f_{\text{dark}} = \mathbb{E} \left[\frac{\sum_x \mathbb{I}[F_t(x) < \tau_d]}{|\Omega|} \right] \quad (25)$$

$$f_{\text{bright}} = \mathbb{E} \left[\frac{\sum_x \mathbb{I}[F_t(x) > \tau_b]}{|\Omega|} \right] \quad (26)$$

$$f_{\text{sharpness}} = \mathbb{E}[\text{Var}(\nabla^2 F_t)] \quad (27)$$

$$f_{\text{corr}} = \mathbb{E}[\text{corr}(F_t^L, F_t^R)] \quad (28)$$

$$f_{\text{texture}} = \mathbb{E}[\|\nabla F_t\|] \quad (29)$$

81 3.8 RPX Difficulty Score

82 All features are percentile-normalized:

$$\tilde{f}_i = \text{PercentileNorm}(f_i)$$

83 The final difficulty score is:

$$\text{RPX-DS} = \sum_i w_i \tilde{f}_i$$

84 Splits are assigned per phase:

- 85 • Easy: bottom 33%
- 86 • Medium: middle 33%
- 87 • Hard: top 34%

88 4 Method

89 Benchmarking tasks

- 90 1. Open World Detection - OWL-ViT Minderer et al. [2023], DITO Kim et al. [2023b], RO-
91 ViT Kim et al. [2023a], YOLO-World Cheng et al. [2024], Sambor Han et al. [2023],
92 GroundingDINO Liu et al. [2024]
- 93 2. Masks
- 94 3. Tracklets - SAM2 Ravi et al. [2024]
- 95 4. Visual Grounding -
- 96 5. Relative Camera Pose Estimation given two image pairs - Mickey Barroso-Laguna et al.
97 [2024]
- 98 6. Absolute depth estimation [Selected pixels] -
- 99 7. Novel-view synthesis - NeRFs, Gaussian Splatting, other new methods
- 100 8. Keypoint correspondences methods - SIFT features Lowe [2004], LoFTR Sun et al. [2021],
101 eLoFTR Wang et al. [2024]

5 Conclusion

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190 Times New Roman is the preferred typeface throughout, and will be selected for you by default.
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194 space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the
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196 For the final version, authors' names are set in boldface, and each name is centered above the
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199 co-author side by side.

200 Please pay special attention to the instructions in Section 9 regarding figures, tables, acknowledgments,
201 and references.

202 **8 Headings: first level**

203 All headings should be lower case (except for first word and proper nouns), flush left, and bold.

204 First-level headings should be in 12-point type.

205 **8.1 Headings: second level**

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207 **8.1.1 Headings: third level**

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211 **9 Citations, figures, tables, references**

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214 The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as
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216 acceptable as long as it is used consistently.

217 The documentation for `natbib` may be found at

218 `http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

219 Of note is the command `\citet`, which produces citations appropriate for use in inline text. For
220 example,

221 `\citet{hasselmo}` investigated\dotso

222 produces

223 Hasselmo, et al. (1995) investigated...

224 If you wish to load the `natbib` package with options, you may add the following before loading the
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229 `\usepackage[nonatbib]{neurips_2026}`

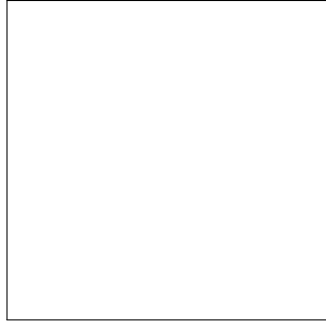


Figure 2: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

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Note that footnotes are properly typeset *after* punctuation marks.²

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All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for the first word and proper nouns); figures are numbered consecutively.

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All tables must be centered, neat, clean, and legible. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for the first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the booktabs package, which allows for typesetting high-quality, professional tables:

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¹Sample of the first footnote.

²As in this example.

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Part		
Name	Description	Size (μm)
Dendrite	Input terminal	≈ 100
Axon	Output terminal	≈ 10
Soma	Cell body	up to 10^6

really shouldn't be using \$\$ anyway; see <https://tex.stackexchange.com/questions/503/why-is-preferable-to> and <https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath> for more information.)

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- `xfig` “patterned” shapes are implemented with bitmap fonts. Use “solid” shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

```
\usepackage{amsfonts}
```

followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

```
\newcommand{\RR}{I\!\!R} %real numbers
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```

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```
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\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (<http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf>)

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